

40.520 Stochastic Models

(1) If X is exponential with rate λ .

RTP; $Y = \lfloor X \rfloor + 1$ is geometric with param $p = 1 - e^{-\lambda}$ where $\lfloor x \rfloor$ is floor funcⁿ.

For $n = 1, 2, \dots$
 $X \sim \text{Exp}(\lambda)$

$$\therefore P(Y=n) = P(n-1 \leq X \leq n)$$

Because of floor function

Using the exponential cdf: $\therefore P(Y=n) = P(X \geq n-1) - P(X \geq n)$

$$P(Y=n) = (e^{-\lambda(n-1)} - e^{-\lambda n})$$

Now, if we set $p = 1 - e^{-\lambda}$
 $\Rightarrow e^{-\lambda} = 1 - p$

$$\therefore P(Y=n) = (1-p)^{n-1} p \Rightarrow p(1-p)^{n-1}$$

Geometric dist.

(2) Stopping conditions:

* Poisson process on $[0, T]$, stop exactly at event before T .

* For stopping time $t \in (S, T]$, winning condition is if and only if there are NO events after that until T .

* Therefore stopping event must be: (i) first event after 's'. (ii) last event before 'T'.

∴ Poisson increments:

$$N(T) - N(S) \sim \text{POISSON}[\lambda(T-S)]$$

$$\therefore P(\text{win}) = P(N(T) - N(S) = 1)$$

$$(i) \therefore P(N=k) = \frac{[\lambda(T-S)]^k}{k!} e^{-\lambda(T-S)}$$

For $k=1$.

$$P(\text{win}) = [\lambda(T-S) e^{-\lambda(T-S)}] \quad (\text{ans.})$$

$$(ii) \frac{\partial P}{\partial s} = \lambda(T-s) \left\{ e^{-\lambda(T-s)} [-\lambda(-1)] \right\} + \lambda(-1) e^{-\lambda(T-s)} \geq 0$$

Extremizing

$$e^{-\lambda(T-s)} [(T-s)\lambda - 1] \geq 0$$

Admits undefined solutions

$$\therefore \begin{cases} (T-s)\lambda - 1 \geq 0 \\ (T-s)\lambda \geq 1 \end{cases}$$

$$\therefore s^* = T - \frac{1}{\lambda} \quad (\text{ans.})$$

$$(a) \cancel{P(\text{win})} P(\text{win}) (s^*) = \lambda \left(T - T + \frac{1}{\lambda} \right) e^{-\lambda(T - T + \frac{1}{\lambda})} = e^{-\lambda \cdot \frac{1}{\lambda}} = e^{-1} = \frac{1}{e}$$

∴ Probability of winning for "optimal" s^* is $\frac{1}{e}$

- (3) → Infection process $N(t) \sim \text{Poisson}(\lambda t)$
 → Detection delay $L \sim \text{Exp}(\mu)$

∴ Detection = Infection + Delay

- (a) $P(\text{detected by } t | s) \rightarrow$ Fixing infection time 's'.

$$\therefore s + L \leq t$$

$$\Rightarrow L \leq t - s$$

∴ Since $L \sim \text{Exp}(\mu)$.

$$\therefore P(L \leq t - s) = 1 - e^{-\mu(t-s)}$$

$$\therefore p(s) = 1 - e^{-\mu(t-s)} \quad (\text{ans})$$

- ⊙ early detection $\rightarrow$ large $(t-s)$ • detection ↑
 ⊙ late detection $\rightarrow$ small $(t-s)$ • missing ↑

- (b) Arbitrary infection is located UNIFORMLY on the interval $[0, t]$.

$$\therefore p = E \left[1 - e^{-\mu(t-s)} \right]$$

for uniform dist:

$$p = \frac{1}{t} \int_0^t [1 - e^{-\mu(t-s)}] ds$$

$$= \frac{1}{t} \left[(t) - \int_0^t ds e^{-\mu(t-s)} \right]$$

$$= 1 - \frac{1}{t} \cdot \frac{e^{-\mu t} - 1}{\mu} = 1 - \frac{e^{-\mu t} - 1}{\mu t}$$

$$p = 1 - \left(\frac{1 - e^{-\mu t}}{\mu t} \right) \quad (\text{ans})$$

(c) Estimating λ .

$N_1(t) \rightarrow \text{detection} \sim \text{Poisson}(\lambda t p)$.

$N_2(t) \rightarrow \text{NOT detecting} \sim \text{Poisson}(\lambda t (1-p))$.

$E[N_1(t)] = \lambda t p$. {detection}.

For estimation we can $E[N_1(t)] = N_1(t)$

$$\therefore N_1(t) = \hat{\lambda} t p.$$

$$\hat{\lambda} = \frac{N_1(t)}{t p} = \frac{N_1(t)}{t \left[1 - \frac{1 - e^{-\mu t}}{\mu t} \right]}$$

$$\hat{\lambda} = \left\{ \frac{\mu t N_1(t)}{\mu t^2 - 1 + e^{-\mu t}} \right\} \quad (\text{ans})$$

(d) we want $E[N_2(t)]$.

$$E[N_2(t)] = \lambda t (1-p)$$

$$E[N_2(t)] \approx \hat{\lambda} t (1-p)$$

$$\therefore E[N_2(t)] = \frac{N_1(t)}{t p} \cdot t (1-p) = \left[N_1 \left(\frac{1-p}{p} \right) \right] \quad (\text{ans})$$

$$p = 1 - \frac{1 - e^{-\mu t}}{\mu t}$$

(A) let $S(t) = \sum_{i=1}^{N(t)} X_i$

where $N(t)$ is a poisson process of rate λ , independent of the iid sequence (X_i) , with $E[X_i] = \mu$, $\text{Var}(X_i) = \sigma^2$.

$$\text{Cov}(N, S) = E[NS] - E[N] \cdot E[S]$$

Our requirement is $\text{Cov}(N(t), S(t))$.

$$E[N(t)] = \lambda t.$$

By Wald's identity: $E[S(t)] = E[N(t)] \cdot E[X] = \lambda t \mu$

(A) $S(t) = N(t)\mu$ + noise $\rightarrow$ fluctuations of X_i .
 $\rightarrow$ To calculate $E[N(t) \cdot S(t)]$:

$$S(t) = \sum_i X_i \Rightarrow E[S(t) | N=n] = n \mu.$$

$$\therefore E[N(t) \cdot S(t) | N=n] = n \cdot n \mu = n^2 \mu.$$

$$\Rightarrow E[N \cdot S] = \mu \cdot E[N^2].$$

For poisson RV: $E[N^2] = \text{Var}(N) + (E[N])^2 = \lambda t + (\lambda t)^2$

$$\therefore E[N \cdot S] = \mu \lambda t (1 + \lambda t).$$

$$\therefore \text{Cov}(N, S) = \mu (\lambda t)^2 + \mu \lambda t (1 + \lambda t).$$

$$= \mu \lambda t [(1 + \lambda t) - (\lambda t)] = \mu \lambda t$$

$$\therefore \boxed{\text{Cov}(N, S) = \mu \lambda t} \text{ (ans)}$$

(5) Emails arrive according to NON HOMOGENEOUS Poisson process with intensity $\lambda(t) = 5 + 2t$.

(a) for NHPP, mean measure on $[a, b]$ is:

$$m(a, b) = \int_a^b \lambda(u) \cdot du.$$

$$\begin{aligned} \rightarrow \text{Mean on } [1, 4] &= \int_1^4 (5 + 2t) dt \\ &= 5(4-1) + (4+1)(4-1). \end{aligned}$$

$$E[N(4) - N(1)] = (4-1)[5 + 5] = \boxed{30}$$

$$(b) \text{ mean} = \int_2^3 (5 + 2t) dt.$$

$$= 5(3-2) + (3+2)(3-2).$$

$$= 5 + 5 = 10.$$

$$\therefore \lambda' = 10.$$

$$\therefore P(N = \lambda') = \text{Poisson}(\lambda') = \left[\frac{e^{-10} \cdot 10}{10!} \right] (\text{ans})$$

$$(c) \text{ Intensity in } [0, 5] = 5 \cdot (5-0) + (5+0)(5-0) = 50.$$

$$\text{Intensity in } [3, 5] = 5 \cdot (5-3) + (5+3)(5-3) = 2 \cdot (5+8) = 26$$

$$P\{N(5) - N(3) \mid N(5) - N(0) = 50\} = \text{binomial} \left(50, \frac{\lambda_t}{\lambda_T} \right).$$

$$\therefore P(N(5) - N(3) > 20 \mid N(5) - N(0) = 50)$$

$$= \sum_{k=20}^{50} \binom{50}{k} \cdot \left(\frac{26}{50}\right)^k \cdot \left(1 - \frac{26}{50}\right)^{50-k} \quad (\text{ans.})$$

(6) Claims arrive as poisson process with rate $\lambda = 5/\text{day}$.

claim amounts are i.i.d, independent of arrivals with $\mu = 100$, $\sigma^2 = 50^2 = 2500$.

$$\text{let } S(t) = \sum_{i=1}^{N(t)} X_i$$

(a) since $N(t) \sim \text{poisson}(5t)$

$$\text{we have } E[N(t)] = 5t \text{ \& } \text{Var}[N(t)] = 5t. \quad \text{claim size}$$

$$\text{Var}[S(t)] = E[N(t)] \cdot \text{Var}(X) + \text{Var}(N(t)) (E[X])^2$$

$$E[S(t)] = E[N(t)] \cdot E[X] \quad \text{\# of claims.}$$

$$= (500t) = (500t)$$

$$\therefore \text{Var}[S(t)] = 5t \cdot 2500 + 5t \cdot 100^2$$

$$= 5t \cdot 100 \cdot (25 + 100)$$

$$= 125 \times 100 \times 5t = 62500t$$

$$E[S(t)] = 500t$$

$$\text{Var}[S(t)] = 62500t$$

(b) Since $N(t) \sim \text{Poisson}(\lambda t)$ → 5t
 $N(1) \sim \text{Poisson}(5)$

$$P(N(1)=0) \sim \text{Poisson}(5)$$

$$\therefore P(\text{no claims on a day}) = e^{-5}$$

(c) Given 10 claims, what is expected total.

$$\text{If } N=10 \rightarrow S(1) | (N(1)=10)$$

$$\therefore S = X_1 + X_2 + X_3 + \dots + X_{10}$$

$$\therefore E[S(1) | N=10] = 10\mu$$

$$= \boxed{1000}$$

(d) Expected capital @ $t=30$ days.

$$C(t) = 10000 - 0.05t - S(t)$$

$$E[C(t)] = 10000 \exp(-0.05t) - E[S]$$

$$\text{@ } t=30$$

$$E[S(30)] = 500 \times 30 = 15000$$

$$\therefore E[C(30)] = 10000 \exp(-0.05 \times 30) - 15000$$

$$= 10000 \exp(-1.5) - 15000$$

$$\approx 2.98 \times 10^4 \approx \boxed{3 \times 10^4}$$

(5)

2 claims/day $\rightarrow p = 1/2$ 6 claims/day $\rightarrow p = 1/6$

$$(a) P(\Lambda = 2) = \frac{1}{2}, \quad P(\Lambda = 6) = \frac{1}{2}$$

$$P(N(t) | \Lambda = \lambda) \sim \text{Poisson}(\lambda t)$$

$$E[N(t)] = E[E[N(t) | \Lambda]] = E[\Lambda t] = t E[\Lambda]$$

$$E[N(t)] = t \left\{ \frac{1}{2} \cdot 2 + \frac{1}{2} \cdot 6 \right\} = 4t$$

$$\therefore E[N(t)] = 4t$$

$$\text{Var}(N) = E[\text{Var}(N | \Lambda)] + \text{Var}(E[N | \Lambda])$$

$$\text{we, } \text{Var}(N | \Lambda = \lambda) = \lambda t$$

$$\Rightarrow E[\text{Var}(N | \Lambda = \lambda)] = E[\lambda t] = 4t$$

again,

$$E[N | \Lambda] = \lambda t$$

$$\Rightarrow \text{Var}[E[N | \Lambda]] = \text{Var}(\lambda t) = t^2 \text{Var}(\Lambda)$$

$$= t^2 \{ E[\Lambda^2] - (E[\Lambda])^2 \}$$

$$= t^2 \left\{ \frac{1}{2} \cdot 2^2 + \frac{1}{2} \cdot 6^2 - \left(\frac{1}{2} \cdot 2 + \frac{1}{2} \cdot 6 \right)^2 \right\}$$

$$= t^2 \{ 2 + 18 - 16 \} = 4t^2$$

$$\therefore \text{Var}[N(t)] = 4t + 4t^2 = 4t(t+1) \quad (\text{ans})$$

$$(b) P(N(A)=0) = \frac{1}{2} e^{-2} + \frac{1}{2} e^{-6}$$

$$P(\text{No claims}) = \approx 0.0689 \approx \boxed{0.07}$$

$$(c) P(\Lambda \geq 6 | N(1) = 4)$$

$$= \frac{P(\Lambda \geq 6) \cdot P(N(1)=4 | \Lambda=6)}{P(N(1)=4)}$$

$$P(N(1)=4 | \Lambda \geq 6) = e^{-6} \cdot \frac{6^4}{4!}$$

$$P(N(1)=4) = P(N(1)=4 | \Lambda=4) P(\Lambda=4) + P(N(1)=4 | \Lambda=6) P(\Lambda=6)$$

$$P(N(1)=4) = \frac{1}{2} \cdot e^{-6} \cdot \frac{6^4}{4!} + \frac{1}{2} \cdot e^{-2} \cdot \frac{2^4}{4!}$$

$$P(\Lambda \geq 6 | N(1)=4) = \frac{\frac{1}{2} \cdot e^{-6} \cdot \frac{6^4}{4!}}{\frac{1}{2} \cdot e^{-6} \cdot \frac{6^4}{4!} + \frac{1}{2} \cdot e^{-2} \cdot \frac{2^4}{4!}}$$

$$\frac{\frac{1}{2} \cdot e^{-6} \cdot \frac{6^4}{4!}}{\frac{1}{2} \cdot e^{-6} \cdot \frac{6^4}{4!} + \frac{1}{2} \cdot e^{-2} \cdot \frac{2^4}{4!}}$$

$$\therefore P(\Lambda \geq 6 | N(1)=4) = \frac{e^{-6} \cdot 6^4}{e^{-6} \cdot 6^4 + e^{-2} \cdot 2^4}$$

$$= \boxed{0.59735} \text{ (ans)}$$

$$(a) E[A | N(1) = 4].$$

Since Λ only takes 2/6.

$$E[A | N(1) = 4] = 2 \cdot P(\Lambda = 2 | N(1) = 4) + 6 \cdot P(\Lambda = 6 | N(1) = 4).$$

$$\text{let } P(\Lambda = 2 | N(1) = 4) = r.$$

$$\therefore P(\Lambda = 6 | N(1) = 4) = 1 - r.$$

$$\begin{aligned} \therefore E[A | N(1) = 4] &= 2r + 6(1-r) \\ &= 2r + 6 - 6r \\ &= 6 - 4r. \end{aligned}$$

~~Now~~ Now we found $r = 0.59735$ in (7c).

$$\begin{aligned} \therefore E[A | N(1) = 4] &= 6 - 4 \cdot (0.59735) \\ &= \boxed{3.61058} \text{ (ans.)} \end{aligned}$$